End Report of Surveillance

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Table of contents

Table of contents 2
1 Introduction 3
2 Final Product Overview 4
2.1 Subsystem function 4
2.2 Final Configuration 4
Node attachments: 6
2.3 Final Technical Specifications 8
2.4 Final Bill of Materials 8
3 Design Development and Final Ass-Built Design 9
3.1 Original Design 9
3.2 Final Ass-Built Design 10
3.3 Differences Between CDR and Final Product 11
4 Alternative Designs and Methods Considered 12
5 Manufacturing and Production 14
5.1 Production Overview 14
5.2 Equipment and Tools 14
5.3 Manufacturing Process 14
5.4 Manufacturing Challenges and Rework 16
6 Quality Control and Non-Conformities 17
6.1 Manufacturing Quality Control 17
6.2 Deviations and Non-Conformities 17
7 Assembly & Integration Process 18
7.1 Assembly Process 19
7.2 Integration with Other Subsystems 19
7.3 Integration Issues 19
8 Testing and Flight Performance 20
9 Problems and Unresolved Issues 20
10 Lessons Learned 20
11 Recommendations for Future Projects 21

1 Introduction

1.1 Context and Purpose

Surveillance is a project that aims to make the hot fires safer and have a video live stream of the hot fires in case something goes wrong, then there would be a video to hopefully aid in finding the problem. The entire system would also be wireless and battery operated.

1.2 Relevant Project Documentation

Document	**Description**	**Link**
PID	Initial project definition	PID
PDR	Preliminary design	PDR
PDR RID List	RID List for preliminary design	
CDR	Critical design	
CDR RID List	RID List for critical design	
Test Report		
Other	Portal Space Nansen SUR guide	SUR guide

1.3 Acronyms

MC - Mission Control
SUR - Surveillance
BB - Big Brother
Node - Refers to the camera and all its components (Raspberry pi + 3D case + batteries). Each node has been given a name. Node 1 is Tom. Node 2 is Bob. Node 3 is Jerry. Node 4 Chipi. Node 5 is Grettir. Each raspberry pi has been labeled with numbers (we figured it would be a little easier to manage them in conversations if they have a name and or a number)
V - Voltage
A - Amps
* For all names of different parts see Fusion files on the Portal Space hub under the SUR - Surveillance folder

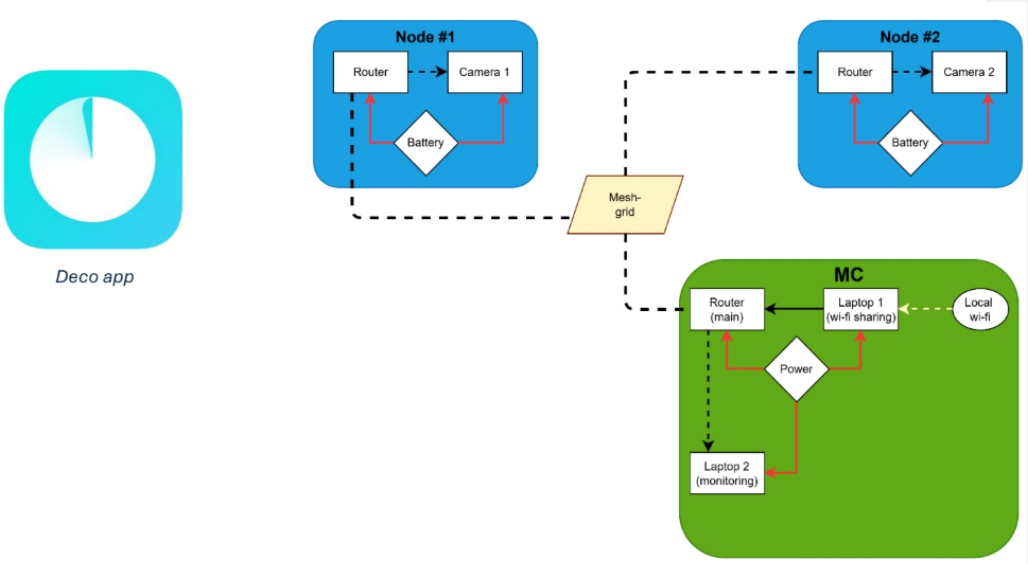
2 Final Product Overview

2.1 Subsystem function

The nodes live stream video through a mesh wifi system and send the live stream to MC. The live stream then gets screen recorded using OBS Studio for later use.

2.2 Final Configuration

WiFi mesh system:
The setup is composed of two parts the mesh-grid and the camera streams. For the mesh-grid, we use Tp-Link Deco wi-fi routers with mesh technology. For the cameras, we use Raspberry Pi 4b. To complete the setup, 2 laptops are required. One laptop is connected to the wi-fi and provides internet sharing connection (ISC) through an ethernet cable between the laptop and the Deco router. The other laptop connects to the mesh and monitors/controls the livestream. See SUR guide for more information.



Topology of the setup. Mission Control (MC) sets up the mesh connection with Laptop 1, while monitoring and activating livestreams happens on Laptop 2

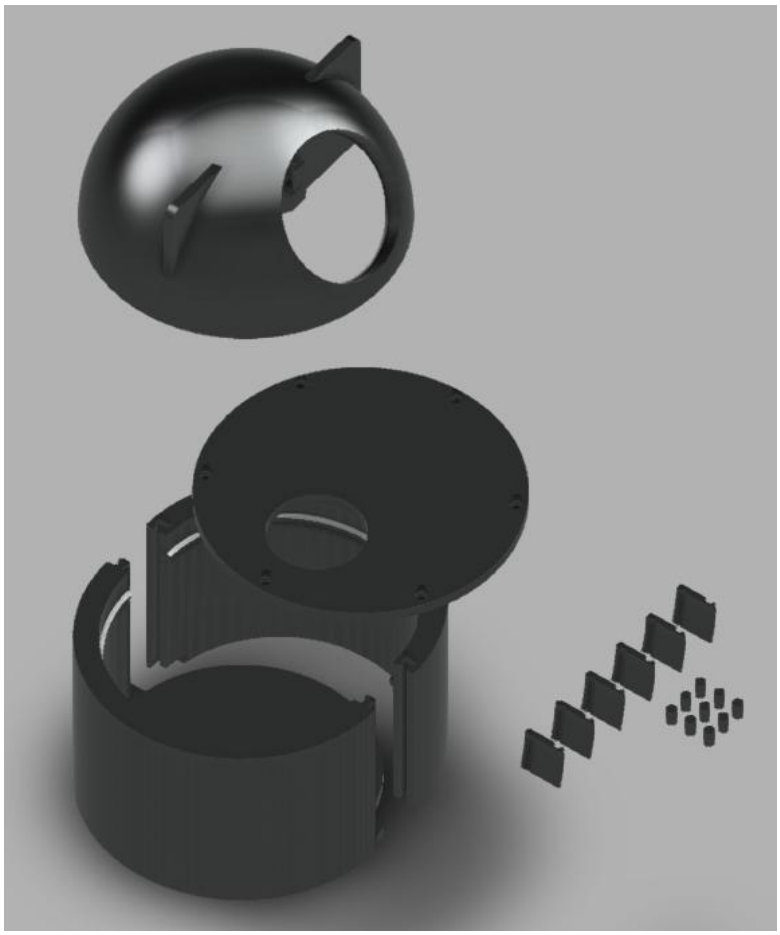
Nodes:
The nodes have a 3D printed case that houses a Raspberry pi 4b (one of them is using a raspberry pi 3 B) and some batteries. Two of them have PiSugar batteries and the rest is using batteries that have been soldered together to create the correct amperage and voltage for the Raspberry pi's.

Routers:
The routers don't have a case as of now but can re-use the old cases left from the BB project but at least two more cases need to be made (3D-printed). The routers have their own battery source so they are completely wireless. The only node that isn't wireless is the one in MC that needs a constant power source (plugged in the wall) and an ethernet connection to a PC.

Node case design:
The design of the node case is pretty simplistic consisting of 4 main parts:

- the top part that holds the camera (Node head)
- the left and right side of the body (arbitrary) where the raspberry pi and batteries are
- and an attachment part that holds everything together (Attachment plate)

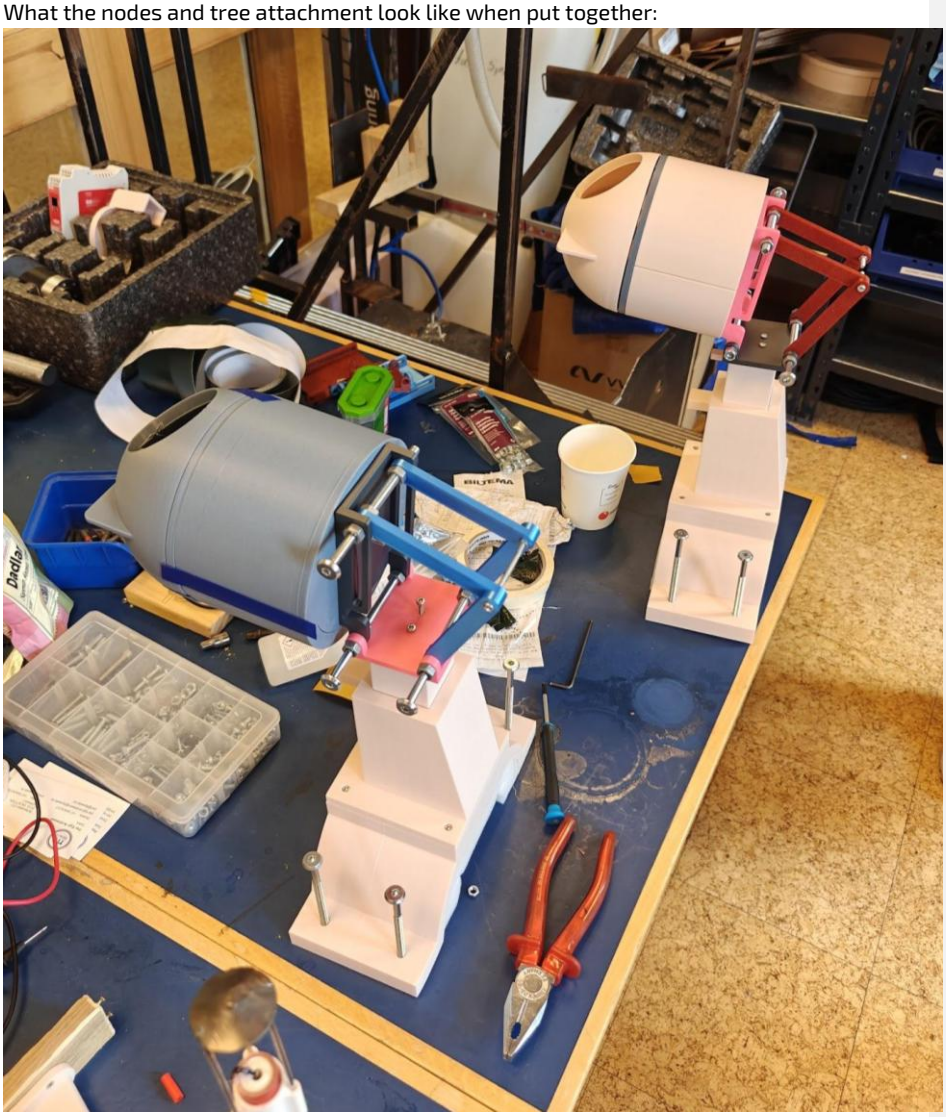
plus 11 smaller parts that need to be glued into the attachment part.
(Exploded model below)



Node attachments:

Tree attachment consists of 3 components, Hinge and Mount + Rail + Clamp that all come together to create the tree attachment so that we can put our nodes on trees.

What the nodes and tree attachment look like when put together:



2.3 Final Technical Specifications

Parameter	Final Value	Unit	Notes/reference
Height of case	15	cm	See Fusion files on PS hub
Diameter of case	13	cm	See Fusion files on PS hub
Mesh range	~125	m	Pictures in the SUR chat on Slack

2.4 Final Bill of Materials

Item No.	Component/ Part	Quantity (in design)	Specifications/ Manufacturer/ Supplier/ Source	Material	Cost per item	Total cost
1	Raspberry pi 4B	3	previous project BB	pcb	free	0 kr
2	Raspberry pi 4B	2	Komplett	pcb	1338,5 kr	2677 kr
3	Raspberry pi 3B	1	previous project BB	pcb	free	0 kr
4	SD cards	2	Clas Ohlson		379.9 kr	759,8 kr
5	Battery for router	2	Elkjøp	3s 5000mAh - 60C - Gens Ace Bashing G-Tech EC5	638 kr	1276 kr
6	Batteries for raspberry pi	2	PiSugar Kitchen	PiSugar S Plus 5000mAh	288.6 kr	577,15 kr
7	Micro HDMI cable	1	Elkjøp		206,1 kr	206,1kr
8	Switching Voltage Regulators microBUCK 4.555V; 12A	5?	Mouser Electronics	pcb	40,25kr	40,25 kr
9	pcb	?	JLCPCB	pcb	40 kr	40 kr
10	TP-link	3	Chlas Ohlson		1699 kr	1699kr
11	Låsemutter M6	1 pack	biltema		44,9 kr	44,9 kr
12	Fjærskive M6	1 pack	biltema		17,9 kr	17,9 kr
13	Sylinder skrue M6	2 packs	biltema		79,9 kr	79,9 kr
14	Foam rolls	2 packs	biltema		159,8 kr	159,8 kr

15	Project total cost	-	-	-	-	7577,8 kr

3 Design Development and Final Ass-Built Design

3.1 Original Design

DR Document Section 4

The original, original design was to have 3 nodes live stream video to MC using radio and have it be completely wireless and battery operated. It would then have 1 additional node for the social media department and that one would only record locally and not stream.

Then we realised radio is expensive so we switched to wireless hdmi streaming but that was also expensive and would require additional things to make the distance of the transmission longer.

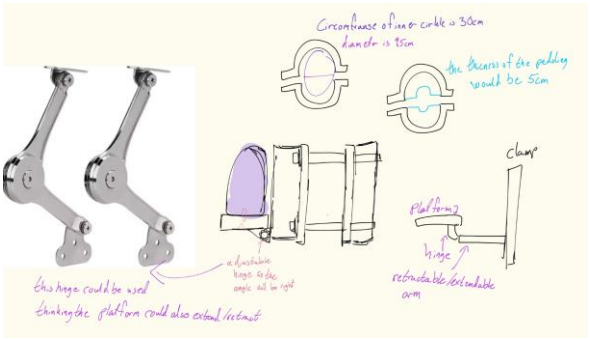
The nodes are supposed to be weatherproof and so we had to come up with some fastening mechanism for the nodes to be weatherproof but still easy to dismantle. The node head was supposed to click into the node body and the node body would slide into each other creating a seal. You would have to slide the node body first in place and then click the head into place. The nodes were originally supposed to be printed in 3 parts. The head, and then two pieces for the node body. The node head stores the camera and the body stores the rest. The node head had a small hole in it for the camera to look out of, the size of the hole was just a filler size until we had gotten the correct measurements for it.

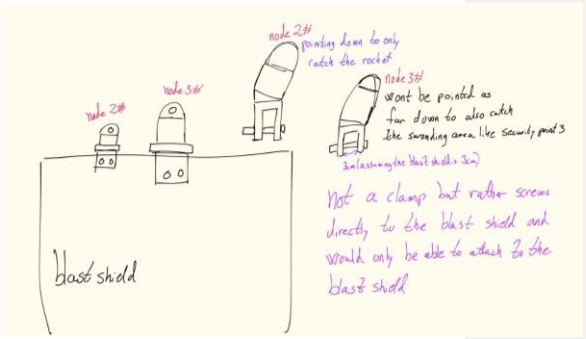
The nodes were originally supposed to store the camera, raspberry pi, and the battery inside itself so the original height was really big to compensate for future possible size of each thing.

We also made space for the possibility that we would have to have some type of PCB that would do something. The use case of that PCB wasn't fully developed because the software and a little of the electronic hardware hadn't been fully realised yet. The PCB was either supposed to store the code for the transmission, or it was supposed to have the HDMI ports, or it was supposed to have the radio in it, or it was supposed to regulate the batteries, it was supposed to be a lot of different things because seemingly we all forgot that we are working with a mini computer and we hadn't nailed down everything when we started.

We wanted the nodes to be attachable to different surfaces during hot fires or launch so we designed it to have a clamp with an extendable arm on its base that can be adjusted for 180 degrees on the end of the arm where the node gets attached. We also had an idea for it to be mounted on the blast shield.

Original design sketches below





3.2 Final Ass-Built Design

The final version of how the nodes transmit the live footage is over a mesh wifi system. See SUR guide for more information.

There are also additional nodes now, originally it was supposed to be just a total of 4 Raspberry pi but at some point two had gotten damaged so two more were ordered but the two that broke got fixed shortly after purchase so we decided to expand the project.

We no longer have one of them record locally for the social media department due to them having it covered with their massive amounts of cameras that they have. Instead we helped them by live streaming one of the hot fires. We were going to live stream the launch too but due to time issues during the launch period SUR was forgotten.

The case of the node is 3D printed and composed of a lot more pieces than originally intended. The ears on the nodes are not optional and need to be there for style and functionality (not actually its just really cute and make them look less inappropriate). We didn't account for how the camera would be held in its position and therefore needed to print the head and then a plate in between the node head and node body (the attachment plate). This plate also keeps the two different parts of the body locked together. The node body is still only two pieces with a sliding mechanism for weather proofing. The body is noticeably smaller due to us realising we don't need a shelf to store the battery and raspberry pi on separate shelves and the batteries and raspberry pi being so small. Making the node smaller, done for the weight and balancing issues it might face on our different attachments.

The routers don't have a specific case for themselves because we figured we could reuse the old ones from BB and hopefully have time to make two more cases for them but we didn't have time for that and they work without a case during good weather.

We dont have a PCB in the normal sense, we just have a PCB for the router's batteries (connecting the battery to the router's power cable). We could make a PCB for the raspberry pi's but for now it's not needed.

We only made one type of attachment (3D printed) and that was the tree attachment. Due to the blast shield never being made and us not knowing how the launch area would look we ended up only making the tree clamp.

3.3 Differences Between CDR and Final Product

The biggest difference between the original design and now is that now we use a mesh wifi system to transmit the video instead of using radio or HDMI. Additional differences is that we have more nodes now and none of them stores the video locally. We figured since they are in the line of fire (literally) we shouldn't have them store it locally since then we need bigger sd cards and if they get damaged the footage is lost so instead we have it recorded on a separate device. The node case is also designed a little differently, for example the height of them is a lot shorter now, and it gets printed in a lot more pieces than originally intended. The hole in the node head is now accurate to the viewing angle of the cameras. The tree clamp hasn't had many changes but rather the idea has been made clearer and the mechanisms have been ironed out. Before the only design we had was how it's supposed to function and hadn't figured out how to make it function.

Item/Area	Critical Design	Final As-Built	Reason for Change	Impact
Video live stream	via HDMI / Radio	via WiFi mesh system	cost effectiveness / budget and lack of knowledge with radio	Parts for the system are less expensive and are easier to replace if needed
3D printed nodes design	3 part system	4 main parts + 11 small parts	quickest and easiest design. A lot smaller now and fits better on a 3D printer bed	More parts that need to be printed and also additional gluing of smaller components to the main part

4 Alternative Designs and Methods Considered

Alternative	Description	Advantages	Disadvantages	Why Not Pursued	Future Relevance
Motion activated camera	have the camera be activated when it detects motion around itself	would in theory save on power usage due to it being in slumber	1. not very useful in this use case. Its only purpose	The camera just isn't in use for long enough for this to be useful	No future relevance

Rejected		when not in use.	isn't just to be a security camera it is also to give constant supervision ability to people in MC or whomever may want to. 2. very expensive and we dont have that big of a budget 3. we would need to make our own software that would run on a computer to make this possible. We would need something that would capture every time it's on and when it's off.	and for it to justify the price we would have to pay for the equipment needed. Would maybe affect our ability to be off grid.	
Movable cameras **Rejected**	cameras that can either manually be moved around to see different angles or can lock on to a target and can follow the target.	when the rocket launches it could follow the rocket	the camera would be really expensive and it was difficult to find a camera that could do it that wasn't already inside a security camera.	it is just an unnecessary function and we made up for it by having the way it's attached to different things be movable so it doesn't serve that big of a purpose. + lack of funding	Could be nice during launch but that's about it. we do have an old camera that can do it so someone would have to figure out how to make it compatible with a raspberry pi
Radio transmition **Rejected**	instead of using the mesh system we have now it would be sending the live footage using radio	The range is so much longer.	The equipment we found was decently expensive and had a big range of price and use cases. The rocket uses radio and so do	lack of knowledge and funding and found out about mesh wifi systems and had everything we needed for it.	could be nice if someone has knowledge of radio, and has a radio license. Could still cause interference with the rocket

			the walkie talkies so there would be a very limited amount of frequencies we could use.		
Using HDMI to transmit the video **Rejected**	instead of radio we could have used wireless HDMI connection	easier to use than radio	EXPENSIVE! would require extenders so would just cost too much and wouldn't offer such a good range	lack of funding and it never got shipped to us and found out about mesh wifi systems and had everything we needed for it.	No future relevance
Reusing old cameras Abandoned	our first plan was to use the camera from old security cameras to have good quality footage and to have some quality of life stuff such as motion detection and movable cameras instead of raspberry pi cameras	a lot of quality of life features we couldn't afford to have. like good quality footage, movable cameras, motion detection. a good base to work on. environmentally friendly	a lot easier said than done. Each system is unique so we would have to figure out each one as it came.	A lack of cameras came so we were behind schedule and so it was scrapped.	We do have them so it's still possible but unnecessary due to the cameras we already have.

5 Manufacturing and Production

5.1 Production Overview

Component/Part	**Manufacturing Method**	**Material**	**Produced By**
Router battery	solder batteries Håkon stole from his job	Batteries - 3.7V 1850mAh 6.85Wh	Ósk and Connor
Raspberry Pi batteries	solder batteries Håkon stole from his job	batteries - 3.7V 1850mAh 6.85Wh	Ósk and Connor
Node case	3D-printed	PLA Basic	Connor

Attachments for nodes	3D-printed	PLA Basic	Connor

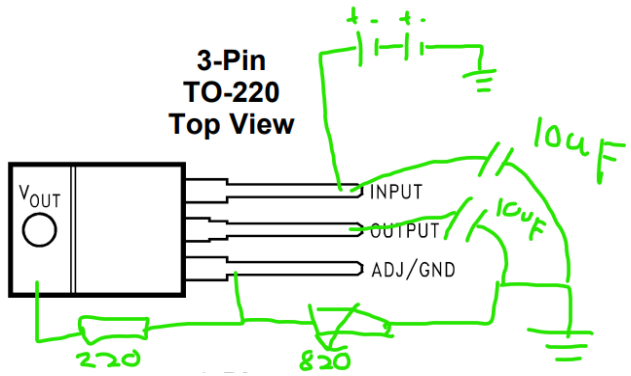
5.2 Equipment and Tools

- Autodesk Fusion
- Bambu Lab X1C printer
- Soldering station
- Raspberry Pi Imager software
- Python
- Deco app

5.3 Manufacturing Process

Home-made router batteries:
We soldered 3 batteries in series. The batteries used were stolen from Håkon workplace (not really stolen since they were going to be thrown away, so the right word is reused). The routers need 12V in order to function properly and the soldered batteries provide 11.1V to the routers which is sufficient.

Home-made raspberry pi batteries:
We soldered 2 batteries in series with a regulator between the battery and the raspberry pi. The batteries used were stolen from Håkon workplace. The raspberry pi's need 5V and 3A and the batteries supply enough of both for the system to work.



Node case:
The node cases are 3D printed using the Bambu Lab X1C printer at VUK016. The settings used are the standard settings for a 0.6 mm nozzle (0.30 mm Standard @BBL X1C 0.6

Commented [1]: This is one of the unsolved problems in surveillance. I would very much like to see a description of your "ultimate" solution for this problem.

Commented [2]: what do you mean

Commented [3]: We have home-made batteries that work along side the ones we bought

Commented [4]: Yes, but they are not rechargable or easy to switch. I would like a clean pcb with inbuilt purpousechoosen regulator and charging circuitry. For example.

Commented [5]: The current solution is currently last minute and not very elegant. For example the batteries for the router run without a regulator, which is not ideal and there is no protection for undercharging which would destroy the batteries if not disconnected before that happens.

Commented [6]: ok i added a paragraph in section 9

nozzle). The Fusion files can be found on the Portal Space hub under the SUR - Surveillance folder, along with a final assembly of the node case.

Attachments for nodes:
Same manufacturing process as for the node bodies and the Fusion files can also be found in the same place.

Raspberry pi's setup:
The raspberry pi's are connected to the camera using a ribbon cable that came with the camera. The camera is placed in a compartment in the node head/lens cap that holds it in place. The battery is then connected to the raspberry pi pins, the pin out can be found on their website.

5.4 Manufacturing Challenges and Rework

Due to some manufacturing challenges, like figuring out how to attach the parts together while wasting the least amount of filament, we decided to change the original design of 3 parts to the current design of 4 main parts and 11 small components (while also keeping the original look). The current design does require a bit more time to glue together all the little tabs and pins into the attachment plate (see Fusion files for names of different parts), while also being careful that you don't put them in crooked. It is an overall simple design that can be improved on further if need be. The current design of the tree attachment is also simple enough with an improvement/redesign that could be made when it comes to the Hinge and Mount on the tree attachment. The extendable arm was a little difficult to design but came along really well. The most difficult part of the entire tree clamp was the angle piece that is used to keep the node at any angle, went through many design processes and was going to be bought but we couldn't find any piece that did what we wanted so we had to make it ourselves.

6 Quality Control and Non-Conformities

6.1 Manufacturing Quality Control

[Document critical inspections and measurements performed on manufactured components. Focus on dimensions, tolerances, surface requirements or characteristics that are important for function or safety.]

Feature/ Check	Method	Expected/ Tolerance	Measured	Pass/ fail	Note
The arm pieces for the Hinge and Mount		Being able to hold up the node	Tried it on	passed	initially failed but then we adjusted how it gets printed and

					watched out for the layerlines and now it passes
Clearance for 3D printed parts	Included clearance in 3D models	0.3 - 0.2 mm clearance	0.25 mm clearance	Pass	

6.2 Deviations and Non-Conformities

Type	Description	Detection stage	Impact	Action taken	Approved by
Node head	The node head stores the camera and is the cameras way of seeing outside	Realised early on that the hole wasn't big enough but we didn't find the viewing angle until much later. Found out that the head needs to be thicker so the pin from the attachment piece wouldn't go through the head.	Did delay the process a bit	Re-worked the piece so the viewing angle is correct and the head is thick enough for the pins to sit securely in place.	-
Node body	The node body (as well as the node case as a whole) went through design changes and tweakings of the originally proposed design	As the system evolved and we knew how and what we were going to use to live stream video		Removed shelf, shrunk it, changed up the design of the interlocking system	-
Tree attachment	This will hold the nodes onto different cylindrical objects.	throughout the making of it	did delay it and was very difficult to think of a solution	Wasn't fully realised so it was developed as we went. We thought of using regular hinges and buying an extendable arm but its surprisingly difficult to find so we had to make our own	-
Blast shield	Two of the nodes were	close to one of the	We made a rough	didnt happen	-

attachment	supposed to be on the blast shield but the blast shield didn't end up happening.	hot fires we found out it hadnt been made yet	design that could be altered upon more information	due to blast shield not being made	
Radio				expensive, lack of knowledge	-
HDMI				expensive and dint arrive	-
PiSugar batteries	The PiSugar batteries were ordered so we could begin testing of the full system once they arrived, the expected time of arrival was 3 weeks from ordering	The batteries ordered did not arrive when they were supposed to	Delays in testing of the full system	Contacted Posten and PiSugar Kitchen numerous times until the problem was solved	-

7 Assembly & Integration Process

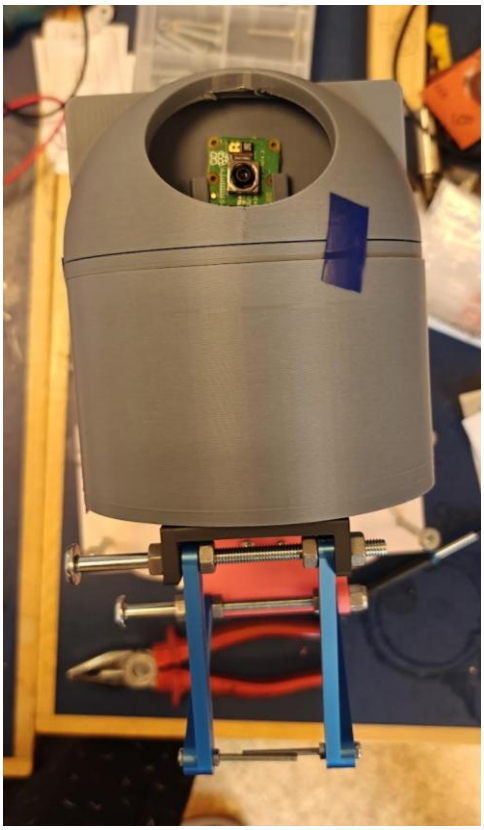
7.1 Assembly Process

The tree attachment is formed of the Clamp, two half circles with foam inside for better grip around different size and textured objects, which would clamp around a tree trunk/branch. From one side it has an extendable rail that can extend to 30 cm, the rail is in two pieces. At the end of the rail is a component (Hinge and Mount) that connects the node case and tree rail + clamp. It has two base pieces, one to be attached to the base of the node body and another to be attached to the end of the rail. The way it works is by having a piece elevated from the base that two bolts can fit through, the bolts can then be moved around to create different angles and then they get tightened to stay in place.

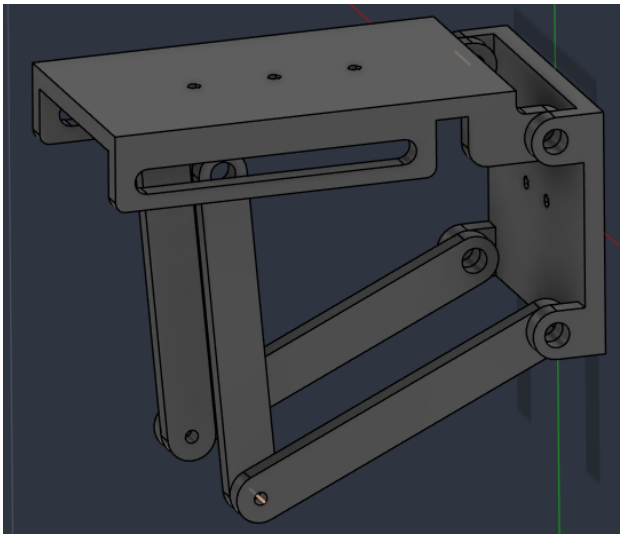
(see picture below)

Attachments for nodes setup:
To make the tree attachment you will need to screw (M3 screws) the Support part (Rail) to the Screw plate (Rail). That will then get screwed to the front part of the Clamp (see Fusion files). Then you screw the first piece of the Hinge and Mount onto the Rail S. You put all the rails inside of the Support (Rail), and lock them in using the little pins. You use the pin to extend the rails to the desired length. Screw the other plate of the Hinge and Mount to the bottom of the node body (M3 screw). You then use three (3) M6 bolts and four (4) M6 nuts for each M6 bolt (!) weave them through the holes of the arms and plates (see picture below on how to do this step - very important).

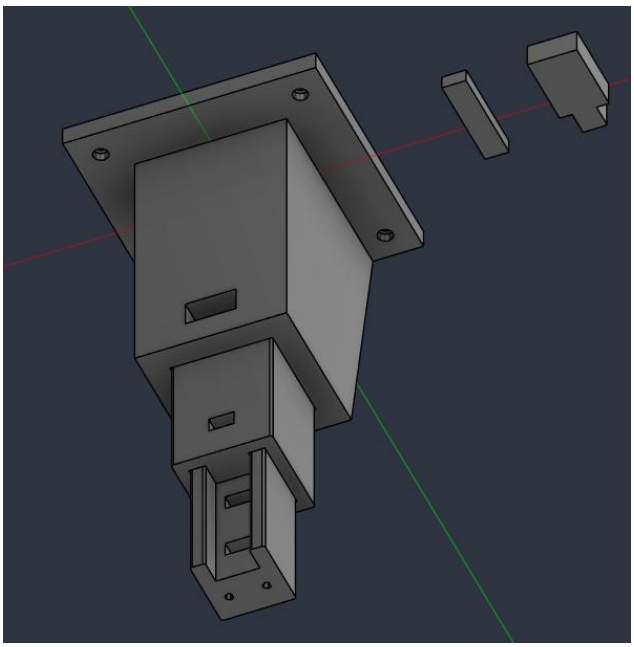
To use the tree clamp you need to put foam on the inside of the clamps so they have a better grip on whatever they are holding and then put each end around the tree or any cylindrical thing and bolt it down using 4 bolts and 4 nuts (M6 bolts and nuts). Two on each side.



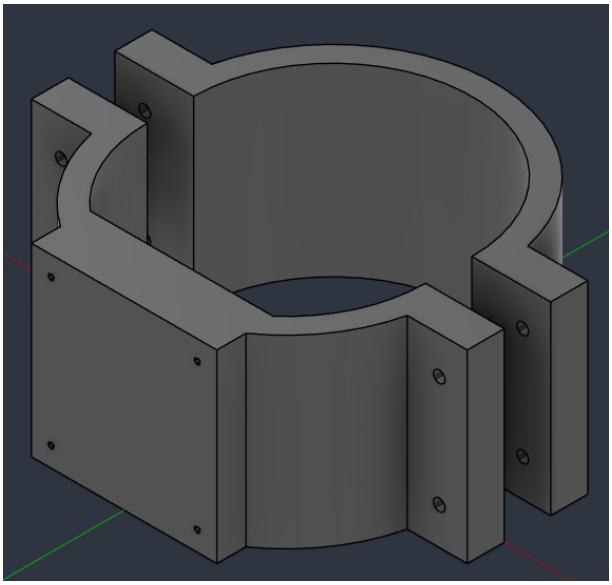
Hinge and Mount:



Rail:



Clamp:



7.2 Integration with Other Subsystems

No integration with other subsystems

7.3 Integration Issues

Some problems occurred with the batteries and the Raspberry Pi camera hardware. The batteries needed to be taken out when it was not used, otherwise it would cause a “leak” which could affect the electronics. So the batteries had to be taken apart after use to avoid this issue.

Some of the cameras experienced a short circuit, and one of the cameras died due to this sensitivity. The cameras that didn’t connect to the mesh as intended had their OS wiped and reinserted with the Raspberry Pi Imager. Both of the cameras managed to connect to the mesh again.

Some issues were discovered regarding the mesh grid. The statuses of the mesh could be monitored through an app (Deco app), however the app at times appeared to be slow. And prematurely disconnecting the mesh, or disconnecting it abruptly could cause

reconnection issues and had to be manually reset. Those issues can be avoided as long as it is set up in the right order.

Another issue that occurred during testing was the pixel colors on some of the cameras. The colors appeared to be reversed. This occurred due to a default setup in the Raspberry Pi OS. The livestream code was modified to account for that.

And the wi-fi hotspot had some challenges regarding which laptop was used as the wi-fi source for the mesh. Using a mac caused some connectivity challenges.

A minor issue occurred with the SD card slots. Some slots didn't connect to the SD card properly, causing the Raspberry Pi Imager to not be able to write/rewrite OS sometimes. New SD card slots were used to fix this issue.

8 Testing and Flight Performance

One of the subsystem components was the mesh grid. The Raspberry Pis were connected to the same mesh in order to access the livefeed. In order to test this, we did several checkpoint tests.

1) Check if one Raspberry Pi was connected to the mesh. To check if it was connected we simply use the Deco app.

2) Check if multiple Raspberry Pi was connected to the mesh. Again, we check if it was connected using the Deco app.

3) Do a distance and obstacle test of the mesh to check how far the mesh points could be, the livestream lag and connectivity issues due to obstacles in the mesh path, like walls, glass, buildings etc.

Another test was the livestreams.

1) Check the livestream of one Raspberry Pi. A laptop (connected to the same mesh) used SSH to connect to the Raspberry Pi and be able to access it remotely. Then we

run the Python script stored in the Raspberry Pi to activate the livestream by generating an URL-link. Accessing this link through a web browser allowed us to verify the livestream.

2) Check livestreams of multiple Raspberry Pi. The same steps were now used to have multiple livestreams at once. Similarly, each camera had to be connected to their respective SSH. Since we had three cameras for the livestream, we would need three separate SSH connections. Each SSH connection runs the Python script and generates a corresponding URL-link. Using three tabs we could now verify the three simultaneous livestreams.

3) Check the livestream lag and quality to see if they meet the acceptable criteria.

To verify the system is working as intended, the Raspberry Pis were tested with batteries instead of a stationary power source.

9 Problems and Unresolved Issues

The major problems were encountered during the full system tests. Raspberry Pis stopped working either because the hardware was failing or SD card was corrupted due to the batteries dying while the raspberry pi was in use. A full test was delayed since we had to wait for new Raspberry Pis. These issues can still occur without us knowing beforehand, making it necessary to have an early system check every time prior to system deployment.

The batteries were a pain. We at first didn't want to buy too many in case they weren't good enough but they took so long to arrive that we didn't have time or the budget to order more so we had to improvise by stealing batteries from Håkon's job and soldering them together. The raspberry pies have a regulator connected to the batteries. The batteries could use improvement but they do technically work.

The node sizes regarding the attachment and mounting needed to be altered when it didn't work or couldn't handle the load.

Another problem would be the home-made batteries (due to them being a last minute solution they're not designed with long term use in mind), which can't be recharged safely since they don't have a regulator (the router ones) and there's also no protection for under/overcharging. Thus making proper PCBs, with a regulator and charging circuitry for the batteries is definitely advised if they're going to be used going forward.

10 Lessons Learned

What happened?

Some cameras died for unknown reasons at first, and despite several tests, we later located the issue, which was the sensitivity of the raspberry pi's.

Another thing was the node components not working properly for the initial stages.

What did we learn?
That electronics hardware can be quite sensitive, and it's important to disconnect the system properly for safety reasons and never take the power source away from the raspberry pi's without turning them off first.

And it's important to consider practical tests of the components like the nodes.

What should be done differently next time?

Include more tests, and that hardware sensitivity needs to be considered next time we test out a system. Not just testing if it works or not, but also somehow checking the robustness and sensitivity of the system. Although in this case it would have been challenging to conduct such sensitivity tests without consultation and sacrificing some Raspberry Pis during the process. In documentation we can add some speculations about how the system will work and what to look out for.

11 Recommendations for Future Projects

What worked well and should be reused?

We set a goal to improve the system by drastically reducing the cables. This also meant that the system was much easier to set up. The mesh grid solution worked very well and can be reused.
Also, the casing for the Raspberry Pi can also be reused.

What should change for the next project?
We believe that minimal changes need to be made to the overall system, with minor tweaks and certain redesigns where previously stated. Perhaps the inventory could add some Raspberry Pi for backup. At most it could benefit from having some fans to cool the system down.

What should future projects investigate?
A possible redesign of certain aspects of the node case (the way all the parts are attached) and the tree attachment hinge and mount part.

A way to extend the range of the wifi routers (but that might require possibly getting a new mesh system?)

The current system has an independent laptop with wi-fi that acts as the hotspot and main source provider for the mesh. One of the routers is connected to the hotspot via an ethernet cable. Perhaps other alternatives can be considered for the hotspot.

A proposed possible alternative for the mesh was to use Local Area Network (LAN). Whether this works (better) with the system and requirements are up for future discussion.

A possible user-experience redesign. Modifying some of the Python scripts or subsystems to activate or connect all the cameras at once instead of doing the same steps several times.

A system display that can resemble a mission control panel would also be nice.

Conduct more tests regarding the system setup. Including different types of laptop and other components.

What should future projects avoid repeating?
Avoid reinventing the wheel, simple is often better (but if it's going to be complicated, then make sure the problem is worth the complexity) :)

Not saying the system can't be improved/upgraded of course

A new surveillance system can be made by just buying a bundle of security cameras on Amazon and using their dedicated app (that is technically more cost effective and easier to use), but that's more operations and not really electronics.

Final words:
Operations made surveillance obsolete. They have better quality video and stream. Surveillance is the most forgotten and useless project out of all the different projects. It's the loneliest project due to the fact that you don't need to talk to any other project and is the most disconnected project out of them all. When you tell other people what you do for Portal Space people don't understand and always say "can't you just buy a security camera?". We did learn something, we learnt how to do mesh WiFi systems and how to 3D model and 3D print. If surveillance will be revived it should be a collaboration between operation and electronics. At the end of the project surveillance just became a part of operations.